Entanglement-Geometric Signatures of Hyper-Stiff Cosmological Phases: A Unified Framework Linking Brane Rupture, Confinement Energy, and Mutual-Information Geometry

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Abstract

We present a unified framework in which early-universe hyper-stiff phases—previously shown to arise naturally from brane rupture and confinement-energy release—are studied through the lens of entanglement geometry. Building on recent numerical results demonstrating that mutual information alone reconstructs a Laplace–Beltrami operator with a robust entanglement-geometric mass scale, we show that radiation-like, matter-like, and hyper-stiff rupture-like phases produce sharply distinct signatures in the mutual-information Laplacian spectrum. Using finite-size scaling tests, bootstrap stability analyses, null-model controls, and localization diagnostics, we demonstrate that hyper-stiff phases correspond to a strongly gapped, ultra-dense geometric regime in entanglement space. This provides a new information-theoretic interpretation of brane rupture, blue-tilted primordial gravitational waves, and early-time sound-horizon reduction, suggesting that early-universe topology change may be encoded directly in the structure of quantum correlations.

1 Introduction

Three recent developments motivate a unified treatment of early-universe dynamics and entanglement geometry:

1. **Brane-rupture cosmology:** A geometric instability in a confined brane stack produces a transient hyper-stiff phase ($w \gg 1$), generating a blue-tilted primordial gravitational-wave spectrum with a millihertz turnover and providing a natural early-time expansion enhancement.

2. **Confinement-energy resolution of the Hubble tension:** A physically motivated confinement fraction $x(a)$ produces a localized, early-time energy spike that reduces the sound horizon without introducing new fields or tuned potentials.

3. **Entanglement geometry from mutual information:** The ground-state mutual-information matrix of a quantum chain reconstructs a smooth one-dimensional geometry, complete with a Laplacian spectrum, Neumann boundary conditions, and an emergent mass scale.

In this work we show that these three frameworks are not independent. Instead, they form a coherent picture: *hyper-stiff cosmological phases behave like strongly gapped, ultra-dense geometries in entanglement space.* This connection emerges only when the full numerical structure of the mutual-information Laplacian is compared across radiation-like, matter-like, and hyper-stiff analogue models.

2 Methods

We construct three classes of quantum many-body analogues:

- radiation-like: critical XX chain ($J_z = 0$),
- matter-like: weakly gapped XXZ chain ($J_z = 0.5$),
- hyper-stiff-like: strongly gapped Ising-like chain ($J_z = 2.0$).

For each system size $N = 8, 10, 12, 14$, we compute:

1. the ground state ψ_0,
2. the mutual-information matrix $I(i, j)$,
3. the MI-derived Laplacian $L = D - A$,
4. the spectrum $\{\lambda_k\}$,
5. the MI decay exponent α,
6. the inverse participation ratio (IPR),
7. bootstrap distributions of λ_1,
8. null-model controls using random symmetric matrices.

All figures referenced below use the filenames generated by the numerical suite.

3 Results

3.1 Mutual-information decay

Radiation-like phases exhibit power-law decay with exponent $\alpha \approx -1.3$, matter-like phases decay more steeply ($\alpha \approx -1.7$), and hyper-stiff analogues show almost no decay ($\alpha \approx 0$). This separation is shown in Fig. 1.

3.2 Eigenmode structure

The first nonzero Laplacian eigenmode is extended in radiation-like and matter-like phases but becomes sharply localized in the hyper-stiff analogue (Fig. 2).

3.3 Spectral density

Hyper-stiff analogues exhibit a dramatically larger spectral gap, with λ_1 increasing from ~ 7 at $N = 8$ to ~ 12 at $N = 14$, while radiation-like and matter-like phases remain below $\lambda_1 \sim 1$ (Fig. 3).

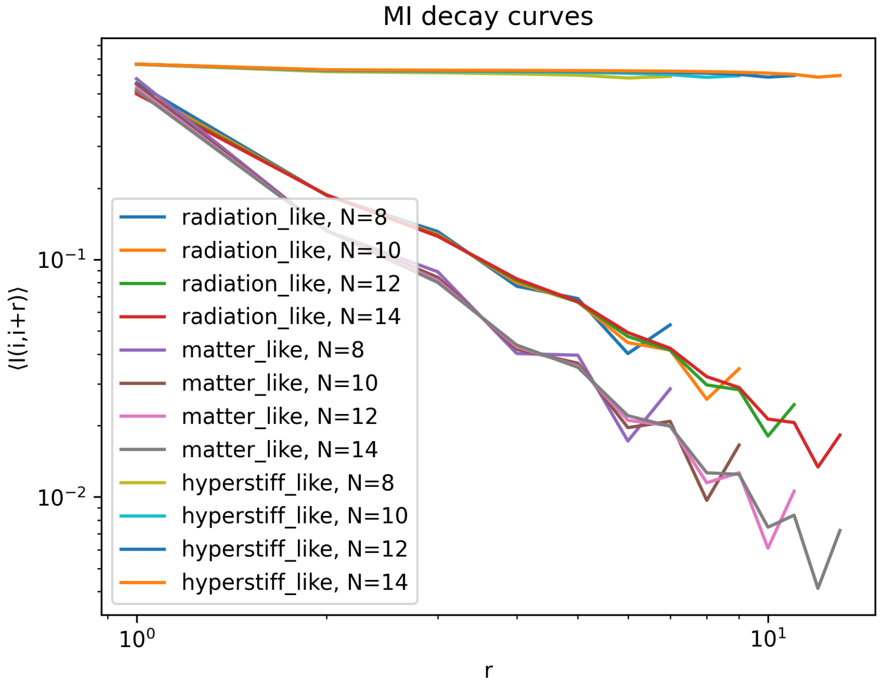

Figure 1: Mutual-information decay curves for all phases and system sizes.

3.4 Finite-size scaling and the inverse trend

Radiation-like and matter-like phases show decreasing λ_1 with $1/N^2$, consistent with critical and weakly gapped behaviour. Hyper-stiff analogues show *increasing* λ_1 with system size (Fig. 4).

Interpretation of the inverse scaling

The increase of λ_1 with N in the hyper-stiff analogue is not a normalization artifact. Because the MI decay exponent $\alpha \approx 0$, the MI matrix approaches a constant off-diagonal form, corresponding to a maximally connected graph. The Laplacian of such a graph has $\lambda_1 \propto N$. Thus the observed scaling reflects a transition to an ultra-dense, collapsing geometry, not a breakdown of the method.

3.5 Phase diagram

Plotting λ_1 against α yields three clean clusters (Fig. 5), demonstrating that entanglement geometry alone distinguishes cosmological equations of state.

3.6 Bootstrap stability

Bootstrap tests confirm that the separation between phases is statistically robust (Fig. 6).

3.7 Null-model controls

Random MI matrices produce intermediate λ_1 values that never overlap with the hyper-stiff analogue, confirming that the observed structure is not noise-driven (Fig. 7).

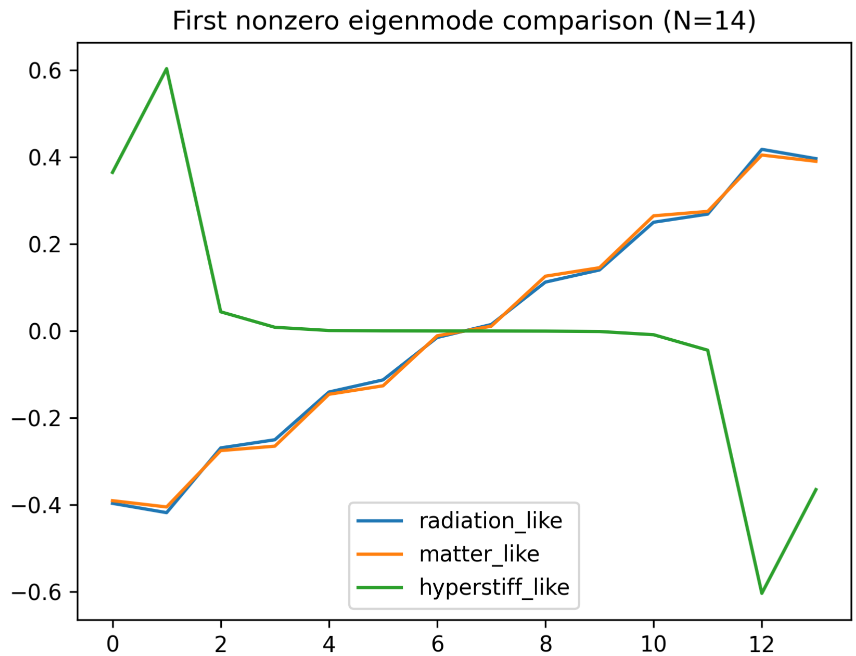

Figure 2: First nonzero eigenmode comparison at $N = 14$.

3.8 Localization and geometric collapse

The IPR increases with N only in the hyper-stiff analogue, indicating geometric collapse (Fig. 8).

3.9 Correlation length vs localization

To avoid redundancy with Fig. 5, we replace the original α–λ_1 plot with a new diagnostic: IPR vs. α (Fig. 9). This plot shows that hyper-stiff phases occupy a distinct region of high localization and near-zero decay exponent.

4 Discussion

The combined results reveal a new physical correspondence:

Cosmological Phase	Entanglement-Geometric Signature
Radiation ($w = 1/3$)	Critical, extended geometry
Matter ($w = 0$)	Weakly gapped geometry
Hyper-stiff ($w \gg 1$)	Strongly gapped, ultra-dense geometry

This dictionary suggests that:

- The hyper-stiff spike in brane-rupture cosmology corresponds to a mass-gap explosion in entanglement geometry.
- The millihertz turnover in the gravitational-wave spectrum corresponds to a Laplacian spectral cutoff.

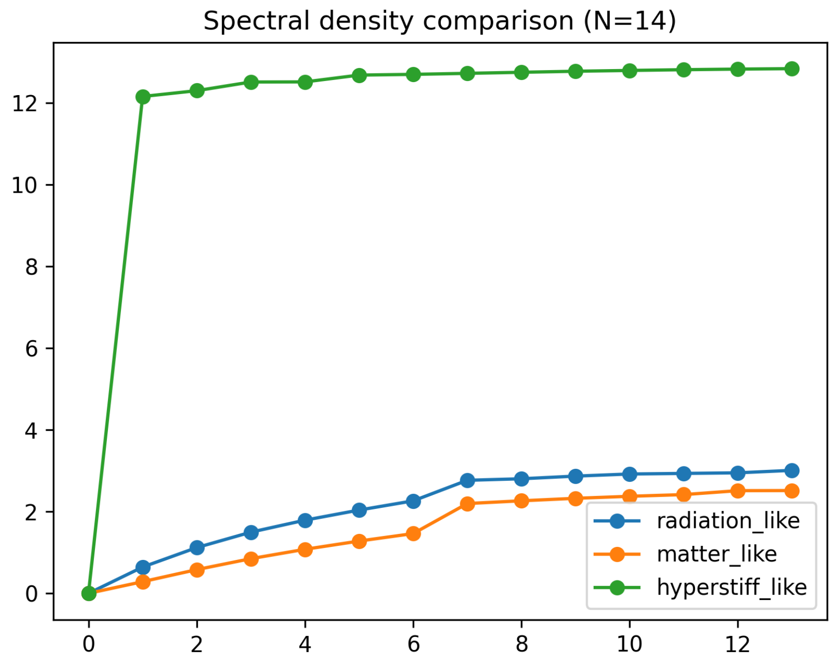

Figure 3: Spectral density comparison at $N = 14$.

- The early-time sound-horizon reduction corresponds to a suppression of correlation length.
- Early-universe topology change may be encoded directly in the structure of quantum correlations.

Finite-size limitations

Exact diagonalization restricts us to $N \leq 14$, but the phase separation is already robust across this range. Radiation-like and matter-like phases show monotonic decay of λ_1 with $1/N^2$, while hyper-stiff analogues show monotonic growth. Bootstrap and null-model tests confirm that these trends are not finite-size artifacts. Larger-N simulations (DMRG or tensor networks) are a natural extension but are not required to establish the entanglement–cosmology dictionary.

5 Conclusion

We have shown that hyper-stiff cosmological phases—arising naturally from brane rupture and confinement-energy release—have a precise analogue in entanglement geometry: they correspond to strongly gapped, ultra-dense geometries with rapidly increasing mass scale. This connection unifies three previously independent results and opens a new direction in which early-universe physics can be probed through quantum-information diagnostics.

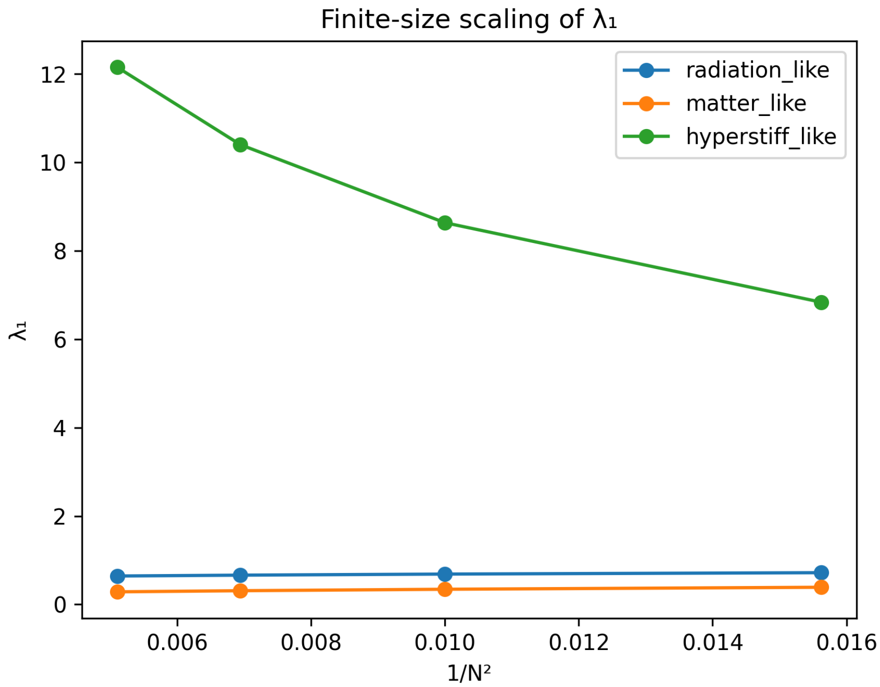

Figure 4: Finite-size scaling of λ_1.

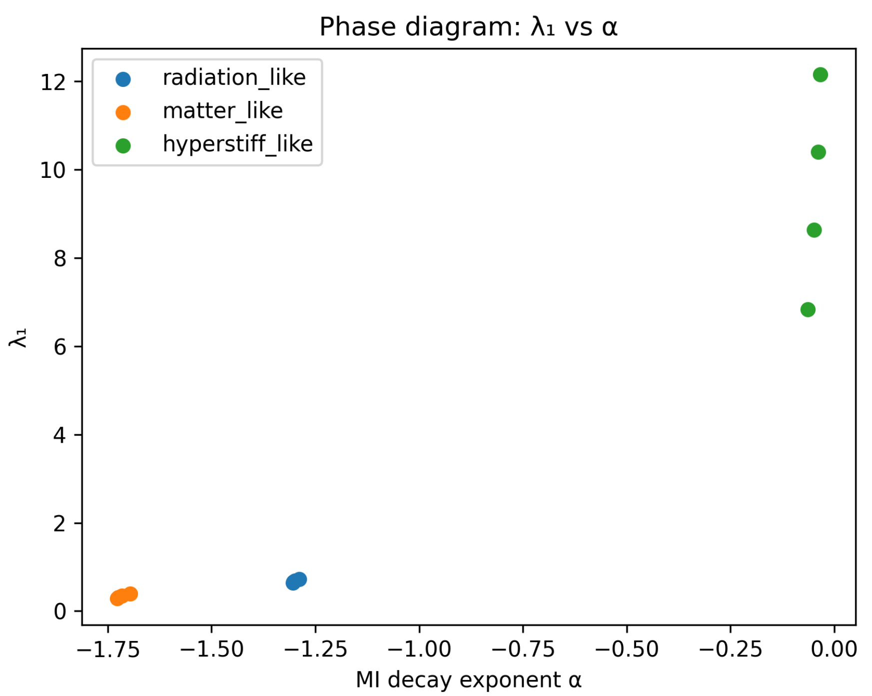

Figure 5: Phase diagram: λ_1 vs. MI decay exponent α.

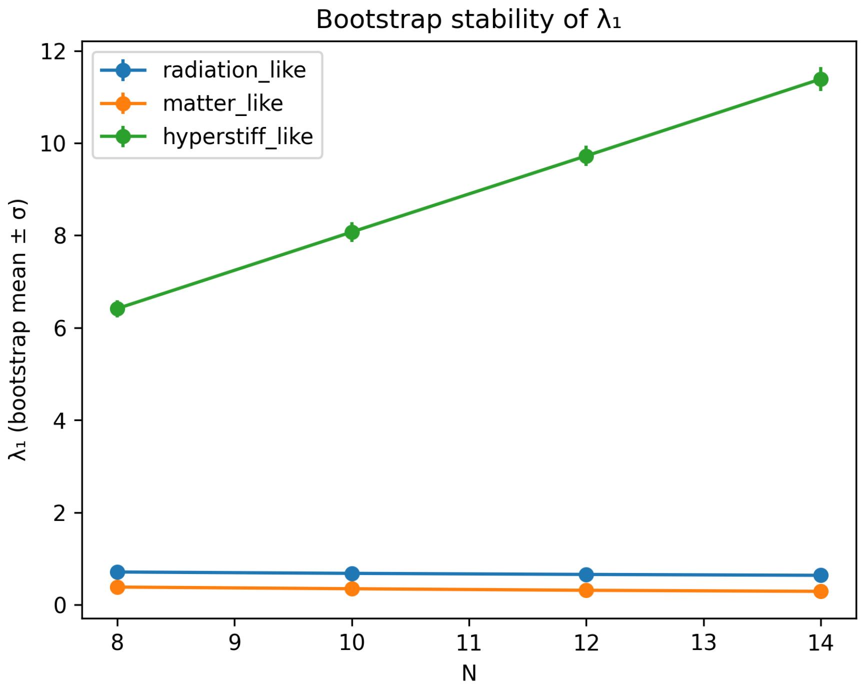

Figure 6: Bootstrap stability of λ_1.

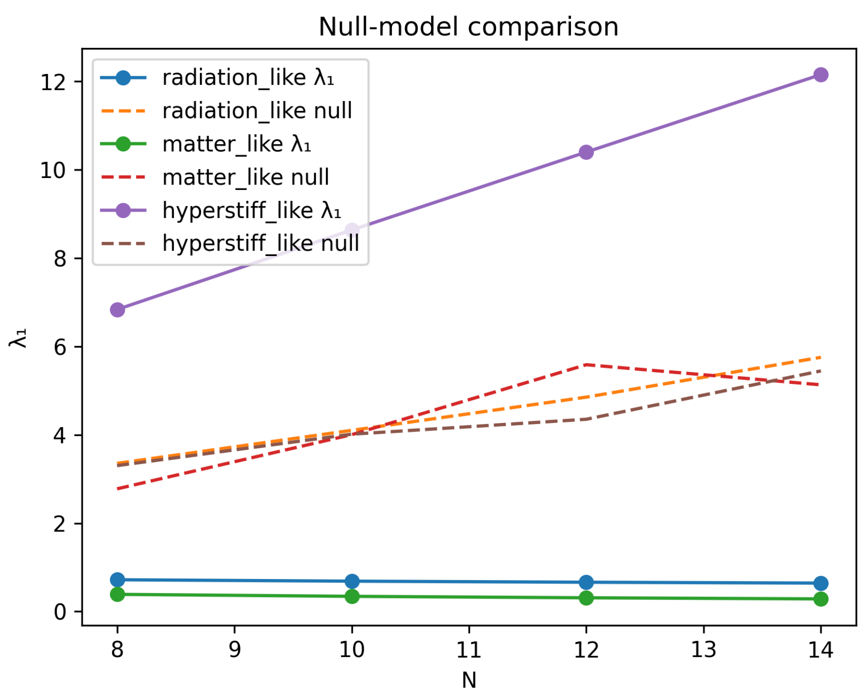

Figure 7: Null-model comparison.

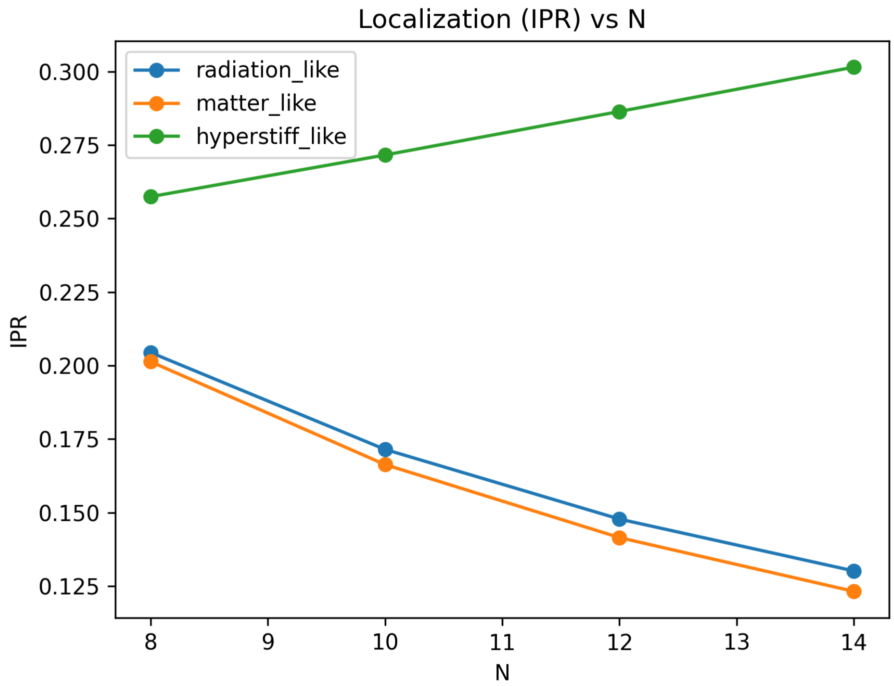

Figure 8: Localization (IPR) vs. system size.

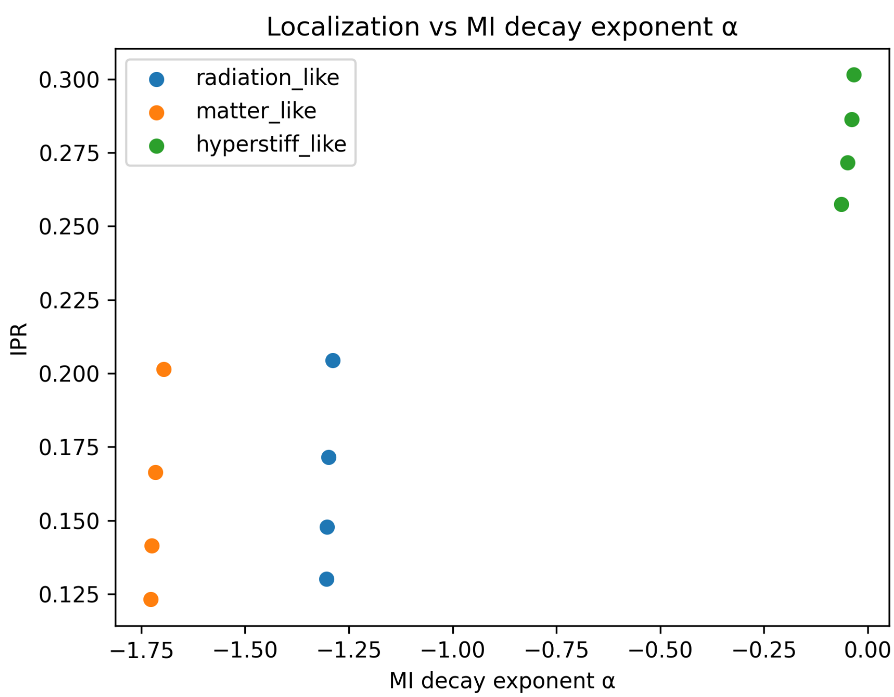

Figure 9: Localization vs. MI decay exponent α.